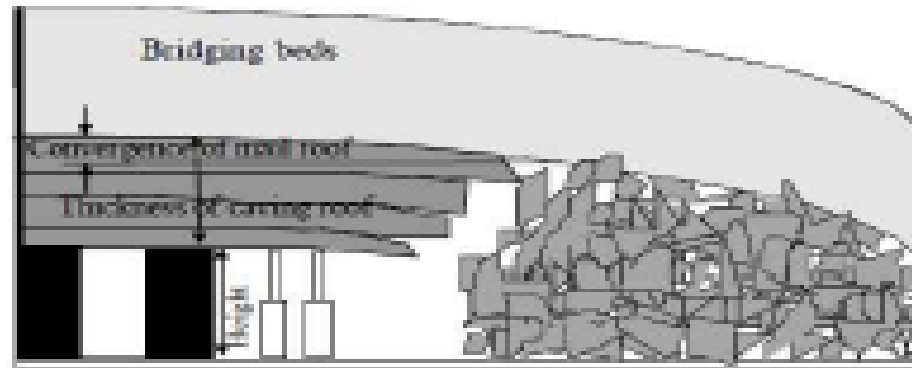


Caving Mechanism

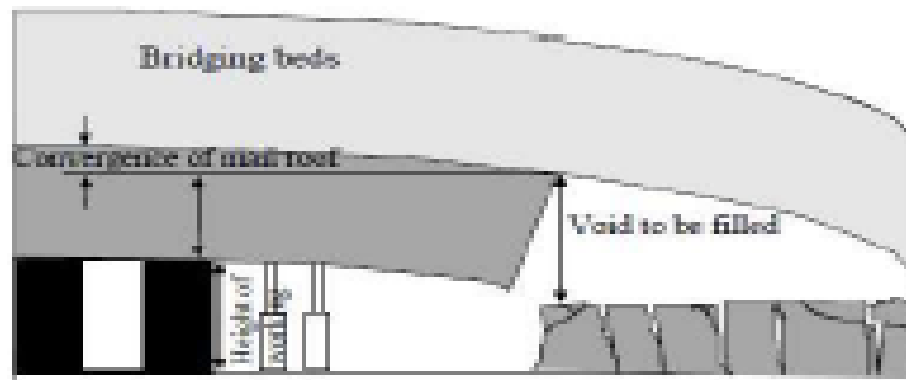
R M Bhattacharjee

Caving Mechanism

Caving Mechanisms – Strata Mechanics



Bulking factor controlled caving of weak and laminated overlying strata.



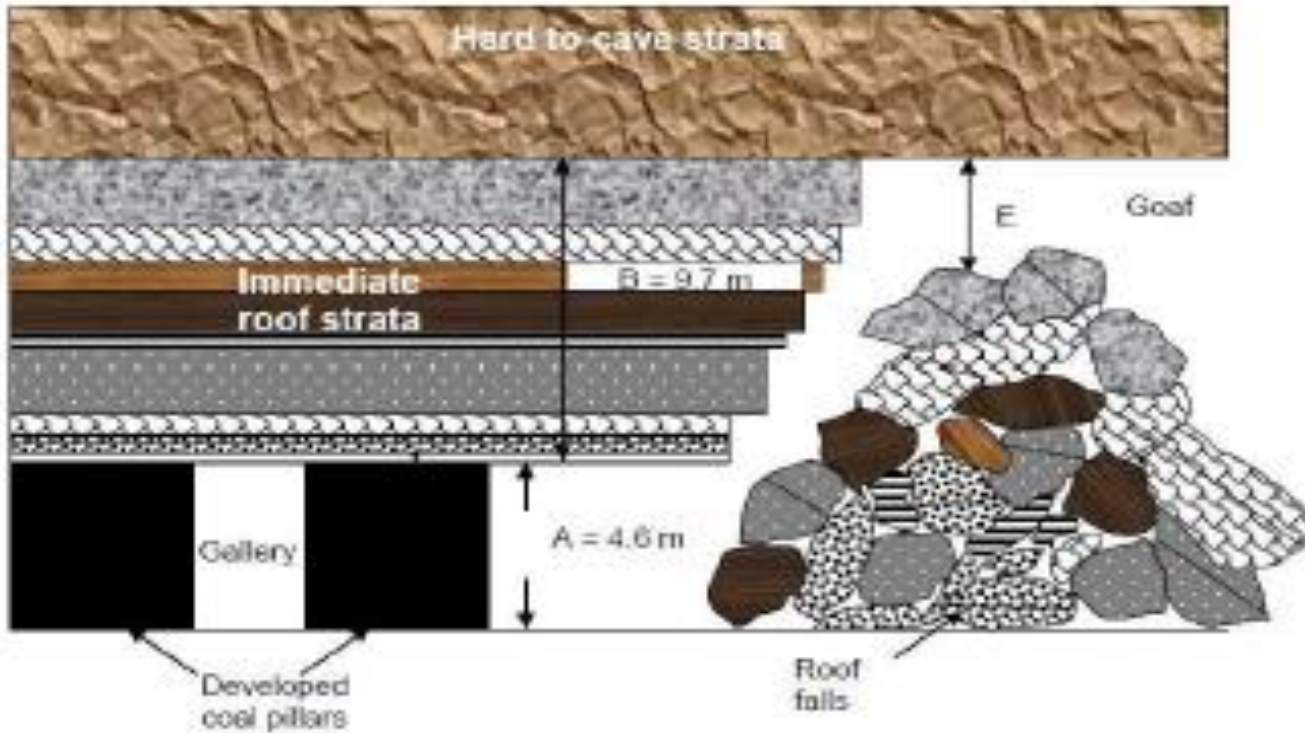
Parting plane controlled caving of strong and massive overlying strata.

For an easily caveable roof stratum, the goaf gets packed quite frequently during face advance. Bulking factor of caved material is important and the face is unlikely to experience dynamic loading.

Working face experiences large overhang if the roof strata are strong and massive in nature. Under this condition, stress meters may play important role to visualise the nature and extent of dynamic loading during enmasse movement of the roof strata.

Caving mechanism

Caving Mechanism in B&P Panels



Cavability

Cavability of a rock formation

Quantitatively, it is difficult to define cavability.

But, a roof may be considered to be ideally cavable when the roof rocks cave in and fill the goaf as soon as the supports are withdrawn.

Caving parameters

Parameters Influencing Caving Span

- **Tensile strength : Measured in laboratory**
- **Rock density : Almost constant**
- **Horizontal stress: Measured in field**
- **Identification of layers and their thickness :
Difficult**

Caving parameters

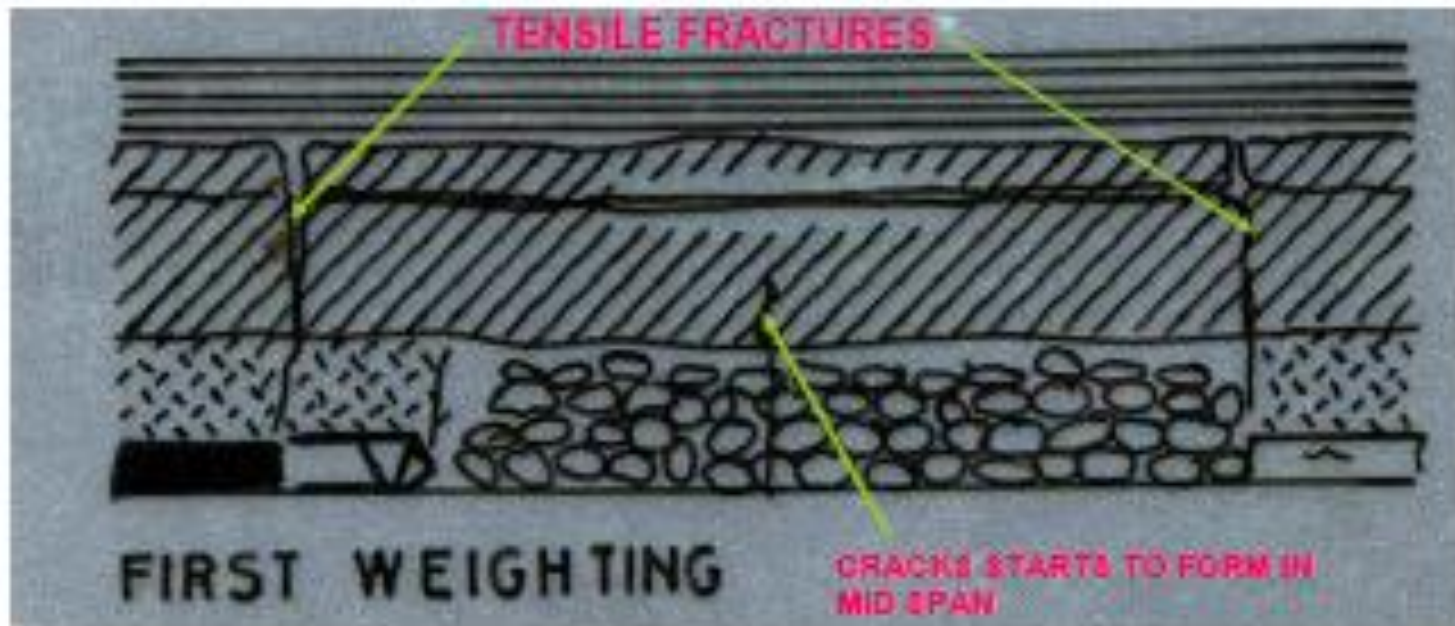
Natural features that influence cavability of rock are

- Geometry of the discontinuities.
- Shear and tensile strength of the discontinuities,
- Strength of rock materials and in-situ stress field

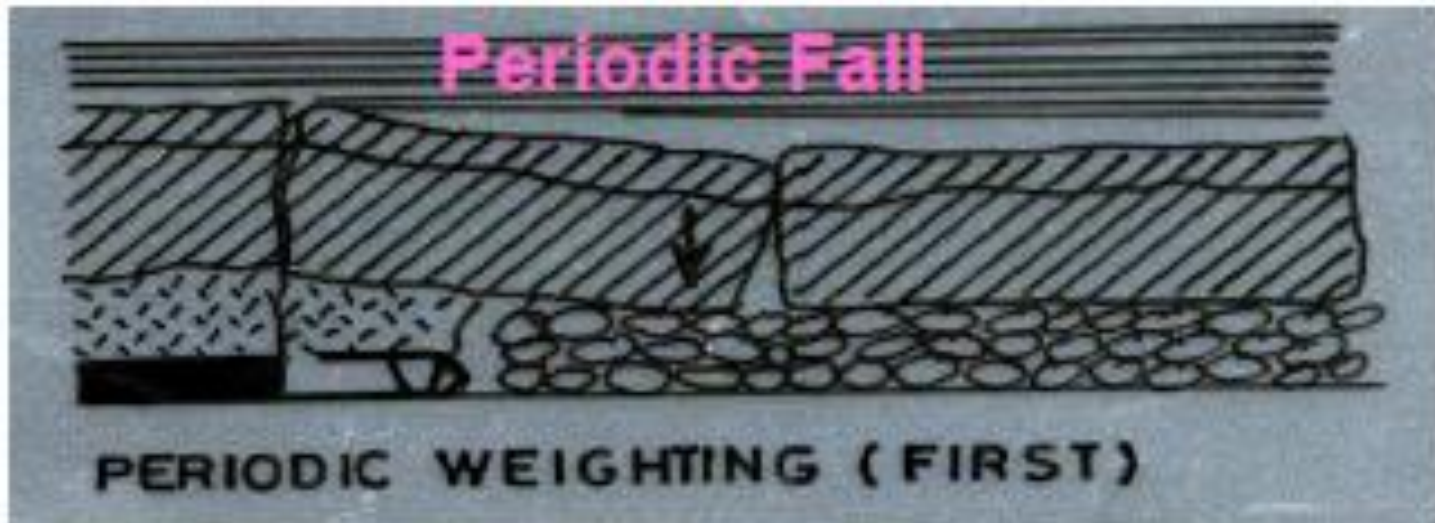
Main Fall

Main Fall

As the retreat further proceeds substantial area of main roof rock forms a plate & caves in by imposing load on supports, known as main weighting.



Periodic weighting



Cavability Index

- Polish scientists (Pawlowicz, 1967) have developed rock quality index, L , to assess the caving behaviour of strata:
- $L = 0.016 C_s d$
- where C_s is the in situ compressive strength of roof rock in kg/cm^2 , and d is the mean discernible thickness of immediate roof strata in cm.

Cavability Index

- The above formula was improved by correlating the in situ strength test result with its uniaxial compressive strength (UCS) test result obtained in laboratory and establishing an empirical relationship between the UCS of roof rock in laboratory and mean discernible thickness of immediate roof (Bilinski and Konopko, 1973). The final equation was proposed as follows:
- $L = 0.064C^{1.7} K_1 K_2 K_3$
where
 - C is the UCS of roof rock measured on dry specimens in laboratory (kg/cm²);
 - K₁ is the in situ strength coefficient, which is 0.33 for sandstone, 0.42 for mudstone, and 0.5 for claystone or siltstone;
 - K₂ is the creep coefficient, which is 0.7 for sandstone and 0.6 for mudstone, clay stone or siltstone;
 - K₃ is the in situ water content coefficient, which is 0.6 for sandstone with 50% relative humidity, 0.4 for clay stone and mudstone with 50% relative humidity.
- Based on the value of L, the roof is categorised in six groups having different values of allowable area of exposure. Good caving of strata is achieved up to a value of L equal to 130 (Class IV roof). A relation has also been established between the span of main fall (S_m) and the roof quality index (L):
- $S_m = 4.47L^{0.4}$

Cavability Index

Mukherjee (2003) used an expression for bending moment of a cantilever to compute the span of failure for cantilever, which simulates the condition of roof failure during periodic weighting at a longwall face:

$$L_p = \sqrt{\frac{\sigma_t t_b}{3\gamma}} \quad (7)$$

where L_p is the span of periodic weighting (m), t_b is the bed thickness (m), and γ is the unit weight of rock (MPa/m).

Cavability Index

Peng and Chiang (1984) proposed a dimensionally correct method of estimating the span of main fall (L_o):

$$L_o = k \sqrt{\frac{h \sigma_{cf}}{\gamma}} \quad (9)$$

where h is the thickness of immediate or main bed; σ_{cf} is the laboratory UCS; γ is the average unit weight of the bed; k is a constant, roughly equal to 0.25. The span of periodic caving was estimated as half the value of main fall span, i.e.

$$L_p = 0.5 L_o \quad (10)$$

Cavability Index

Central Institute of Mining and Fuel Research (CIMFR, erstwhile CMRI) of India proposed an empirical and statistical approach to assess the cavability of strata and support rating (Sarkar, 1998). The cavability of the strata is assessed in terms of caving index, I , of the strongest bed existing within the active caving zone:

$$I = \frac{\sigma_c L_c^n t_b^{0.5}}{5} \quad (11)$$

where σ_c is the UCS in kg/cm^2 , L_c is the average length of core in cm, and n is the constant depending upon the rock quality designation (RQD) of the bed ($1 \leq n \leq 1.2$).

The caving nature of the roof is classified in five groups depending on the value of caving index, I , of the strongest bed (Table 2).

The approach also estimates the spans of main fall (L_o) and periodic caving (L_p) as follows:

$$L_o = 0.72I^{0.51} \quad (12)$$

$$L_p = 3.05 + 0.25L_o \quad (13)$$

Cavability Index

Caving index vs. caving behaviour of strata in longwall faces.

Roof category	Cavability index	Caving nature
I	$I \leq 2000$	Easily cavable roof
II	$2000 < I \leq 5000$	Moderately cavable roof
III	$5000 < I \leq 10,000$	Roof cavable with difficulty
IV	$10,000 < I \leq 14,000$	Cavable with substantial difficulty
V	$I > 14,000$	Cavable with extreme difficulty

Support estimation

The CIMFR of India proposed an empirical and statistical approach to assess the cavability of strata and support rating (Sarkar, 1998). It correlates the maximum face convergence with the caving index of the strongest bed and the thickness of cavable bed between the coal seam and the strongest bed having the highest value of caving index, I . The projected relation is given as

$$C_m = \frac{A}{P} + 9.6h + \frac{KI}{K' + 1.5} - 23 \quad (26)$$

where P is the mean load density (t/m^2); I is the caving index of the strongest bed; A is a constant depending on rock type, which is 1440 for categories I and II, 1700 for categories III and IV, and 1900 for category V rocks; K' is a factor depending on the ratio of thickness of cavable bed between the strongest bed causing the weighting and the coal seam to the extraction height, $K' = 2$ for ratio up to 2, $K' = 3$ for ratio between 2 and 4, $K' = 5$ for ratio between 4 and 8, and $K' = 10$ for ratio above 8; K is 0.025 for sandstone.

Support estimation

Based on the field observation of face convergence and visual observation observed over hundreds of working cycles in several longwall faces in India, a correlation has been established between the face convergence slope and the degradation of roof at the face as given in Table 3.

With the above consideration, the support resistance P is obtained using Eq. (26), such that the corresponding face convergence is acceptable for safe longwall operation, which is taken as 60 mm/m of face advance.

Table 3

Relation between convergence and roof condition (after Sarkar (1998)).

Range of peak convergence (mm/m)	Roof condition
<60	Continuity of roof remains intact with no prominent fracturing
60–100	Minor cracks and breaks and sometimes disjointed blocks are present
100–160	At lower values, prominent fractures are only observed, and rock falls may start occurring at higher ranges
>160	Rock fall causing collapse of the face

Indications of roof weighting

- Increased load on supports
- Cracking sound in goaf
- Face slabbings, rib spalling
- Increase in rate of convergence
- Closure of face heights
- Floor heaving
- Roof deterioration

Precautions against air-blast

- Determine the potential pathways, air velocity, and secondary effects in the vicinity of persons or infrastructure should an airblast occur.
- Consider any effect on ingress and egress from a mine should an air blast occur in various locations.
- Consider planning for safe haven should the potential risk level of an air blast become unacceptable as mining progresses.
- Restrict access ways to potential air blast locations.

Precautions against air-blast

- Monitor rate of convergence and determine withdrawal limit
- Consider establishing bulkheads or rock barriers as a precaution to protect persons and/or infrastructure.
- Provisions for safety valves against air blast
- A dead end cross cut or heading could provide a safe haven in the event of an airblast if there is any warning that one is imminent.
- Personnel should be fully aware of any signs that might indicate an airblast is possible and be fully trained in what action to take if one appears to be imminent.

Precautions against air-blast

- Consider planning good housekeeping to minimise secondary impacts from an airblast from loose material left along drives (or drifts).
- Consider developing a model of potential airblast situations to predict velocities and potential risk to personnel and/or infrastructure.
- Ensure that any installed services adjacent to, or at, the planned breakthrough position are either removed or effectively isolated.
- Mock rehearsals for air blast